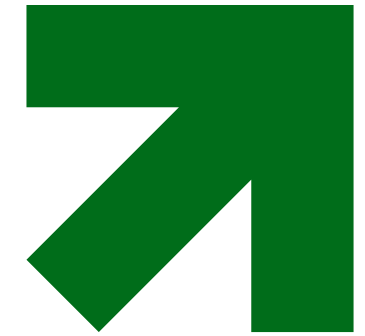




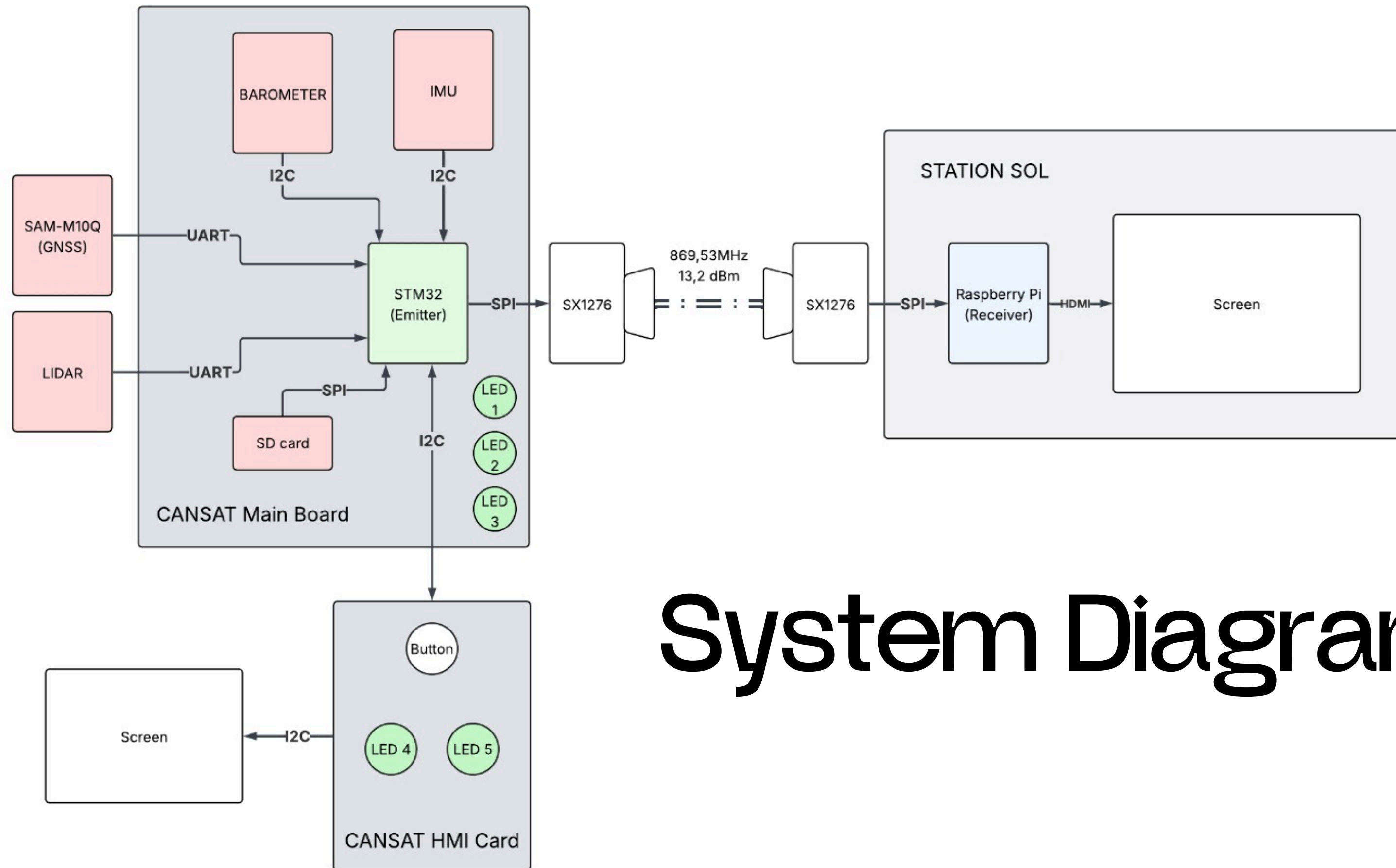
CanSat RCE 2 2025-2026

Alexei DOUILLARD
Juan Pablo BARONA CIFUENTES
Ted KOYAZANDE
Amr TAOUIS
Abdelmoughit HAJJI LAAMOURI

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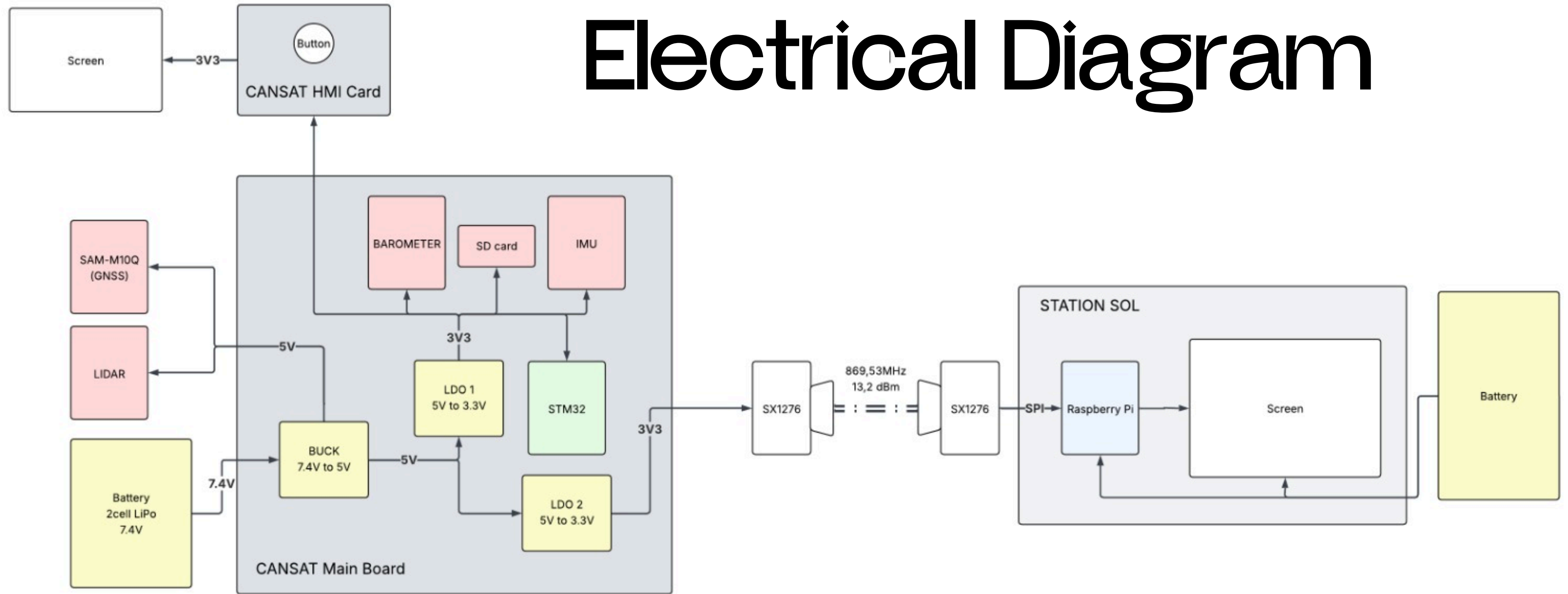


1. Branding
2. Diagrams
3. Progrès du Composants
4. Progrès du modelisation 3D Soufflerie
5. Progrès de la station au sol
6. Progrès du Hardware



System Diagram

Electrical Diagram



Composants

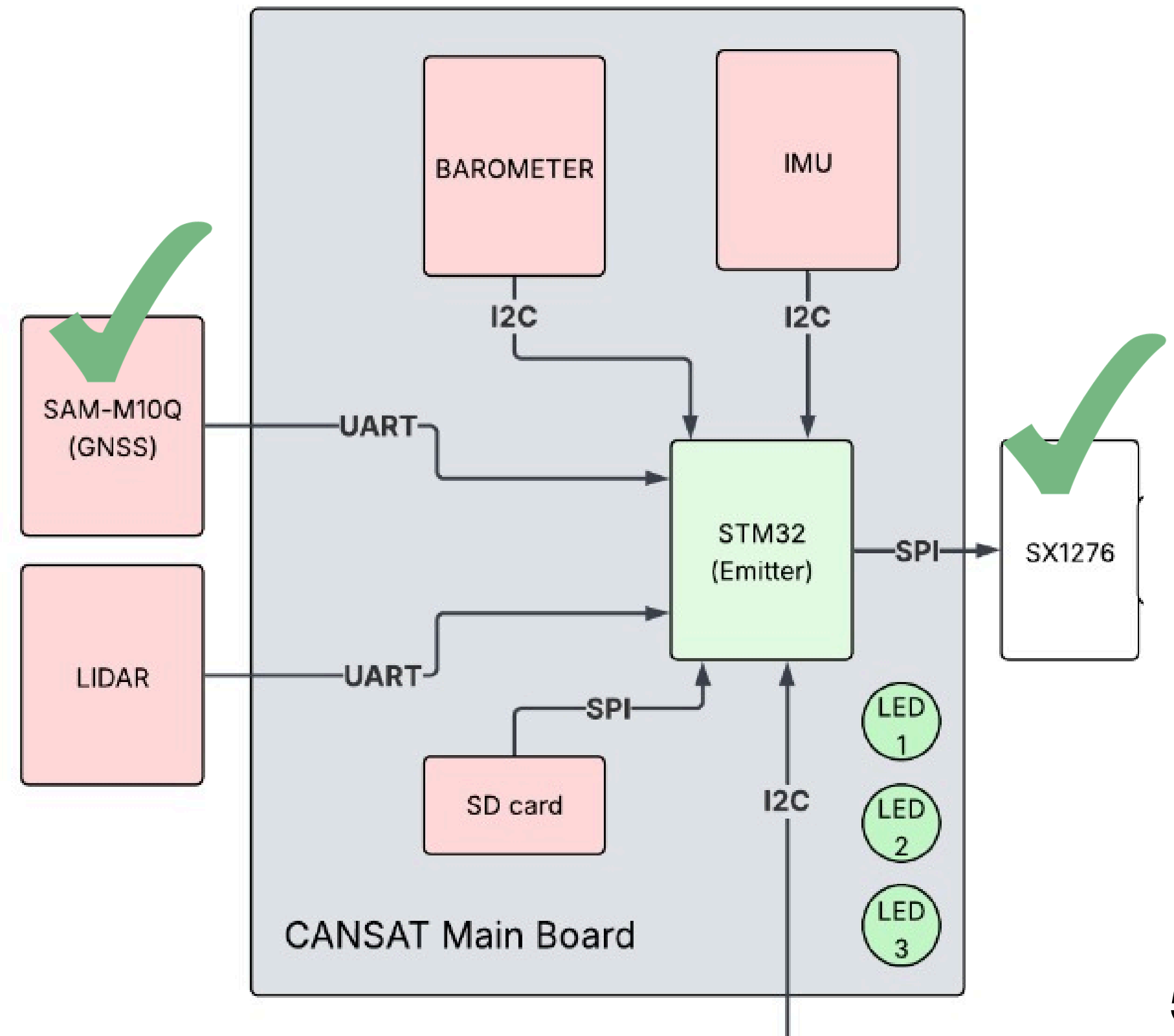


Composant testé individuellement



Composant testé avec notre PCB

état au Cansat Report 4 :

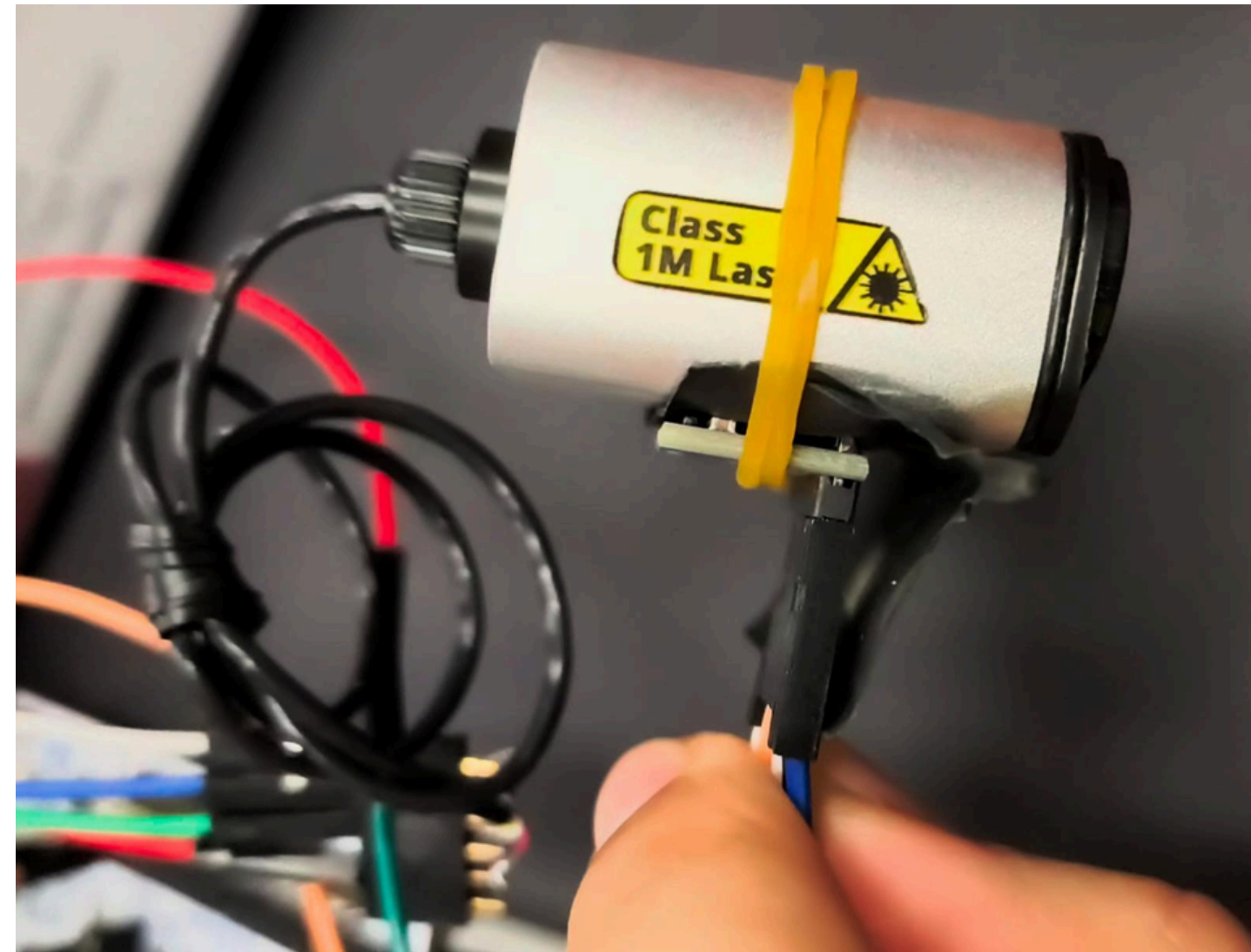


Composants

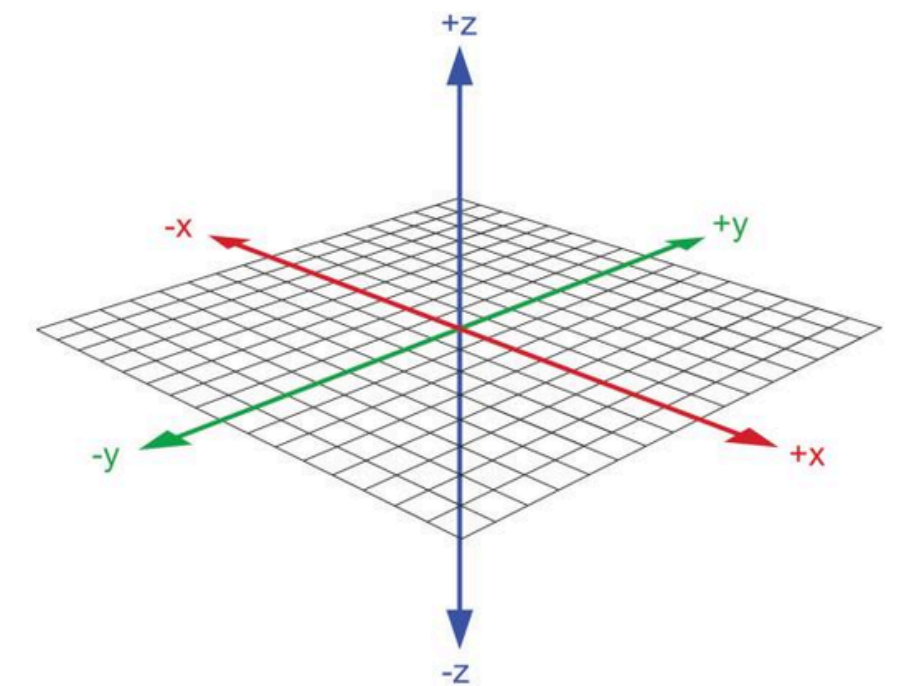
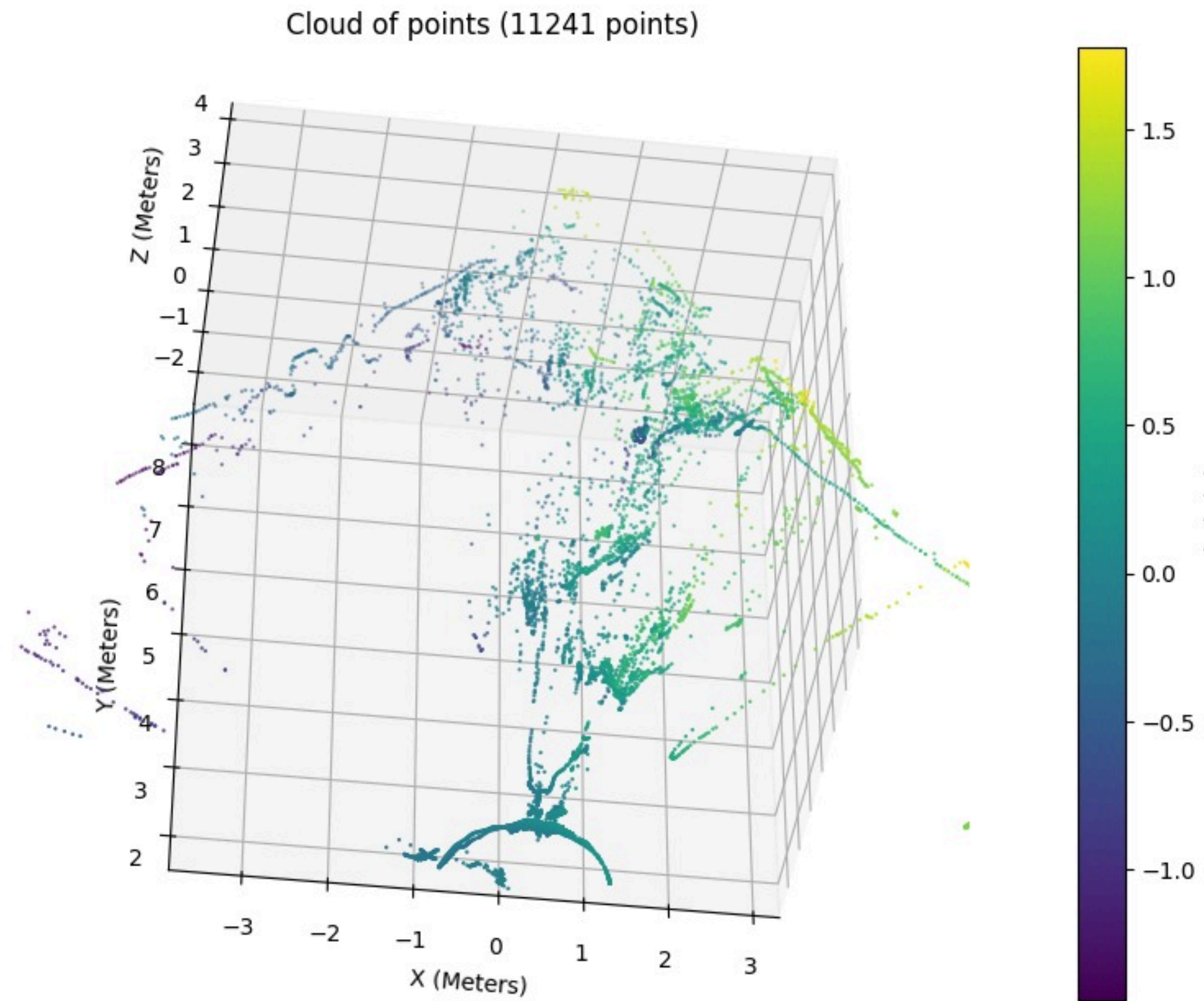
Lidar

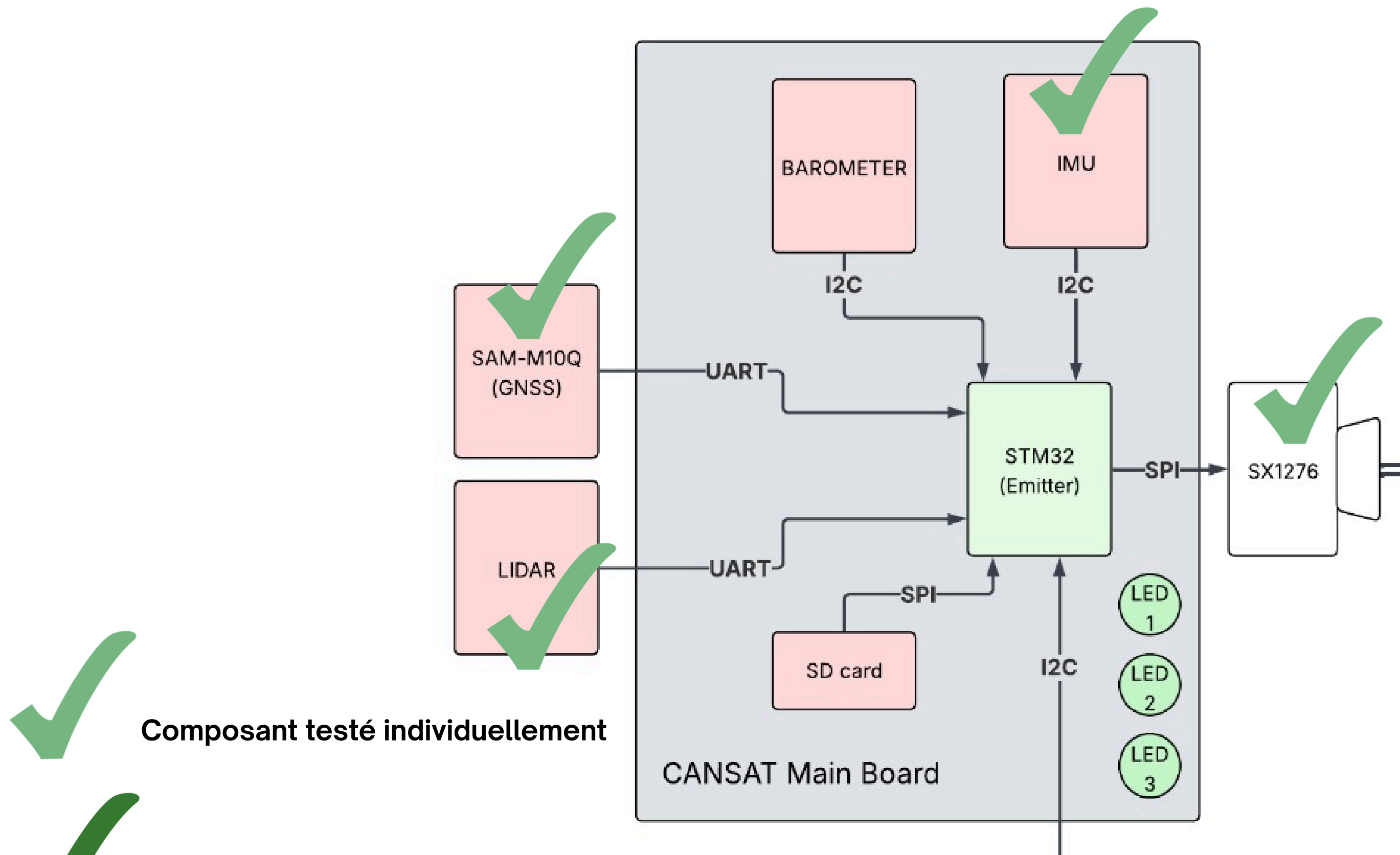
STM32

IMU



400 p/s
115200





Prise en main de la carte SD :

```
145 printf("\r\n==== SD Card ==== \r\n");
146
147 FRESULT res = f_mount(&fs, "", 1); // Necessary before everything else
148 if (FR_OK != res)
149 {
150     printf("Error mounting SD Card\r\n");
151 }
152
153 sd_ls("/");
154
155 FIL f;
156 f_open(&f, "test.txt", FA_READ | FA_WRITE | FA_CREATE_ALWAYS);
157
158 char buf[20];
159 char write_buf[] = "Hello World\r\n";
160 char empty_buf[] = "Empty\r\n";
161 UINT br;
162 printf("Here !\r\n");
163 f_lseek(&f, 0);
164 f_write(&f, empty_buf, 20, &br);
165 f_lseek(&f, 0);
166 f_read(&f, buf, 20, &br);
167 printf("%s\r\n", buf);
168 f_lseek(&f, 0);
169 f_write(&f, write_buf, 20, &br);
170 f_lseek(&f, 0);
171 f_read(&f, buf, 20, &br);
172
173 printf("%s\r\n", buf);
174
175 /* Infinite loop */
176 for(;;)
177 {
178     osDelay(1);
179 }
180 /* USER CODE END StartDefaultTask */
181 }
182
183 /* Private application code -----*/
184 /* USER CODE BEGIN Application */
185
186 /* USER CODE END Application */
187
```

```
# FreeRTOS
# task.h
# main.h
# cmsis_core.h
# stdio.h
# fatfs.h
# defaultT
# defaultT
# StartDe
# MX_FRE
# MX_FRE
# fs:FATF
# sd_ls(co
# StartDe

Here !
Hello World

==== SD Card ====
TEST.TXT
Here !
Hello World

==== SD Card ====
TEST.TXT
Here !
Hello World

==== SD Card ====
TEST.TXT
Here !
Empty

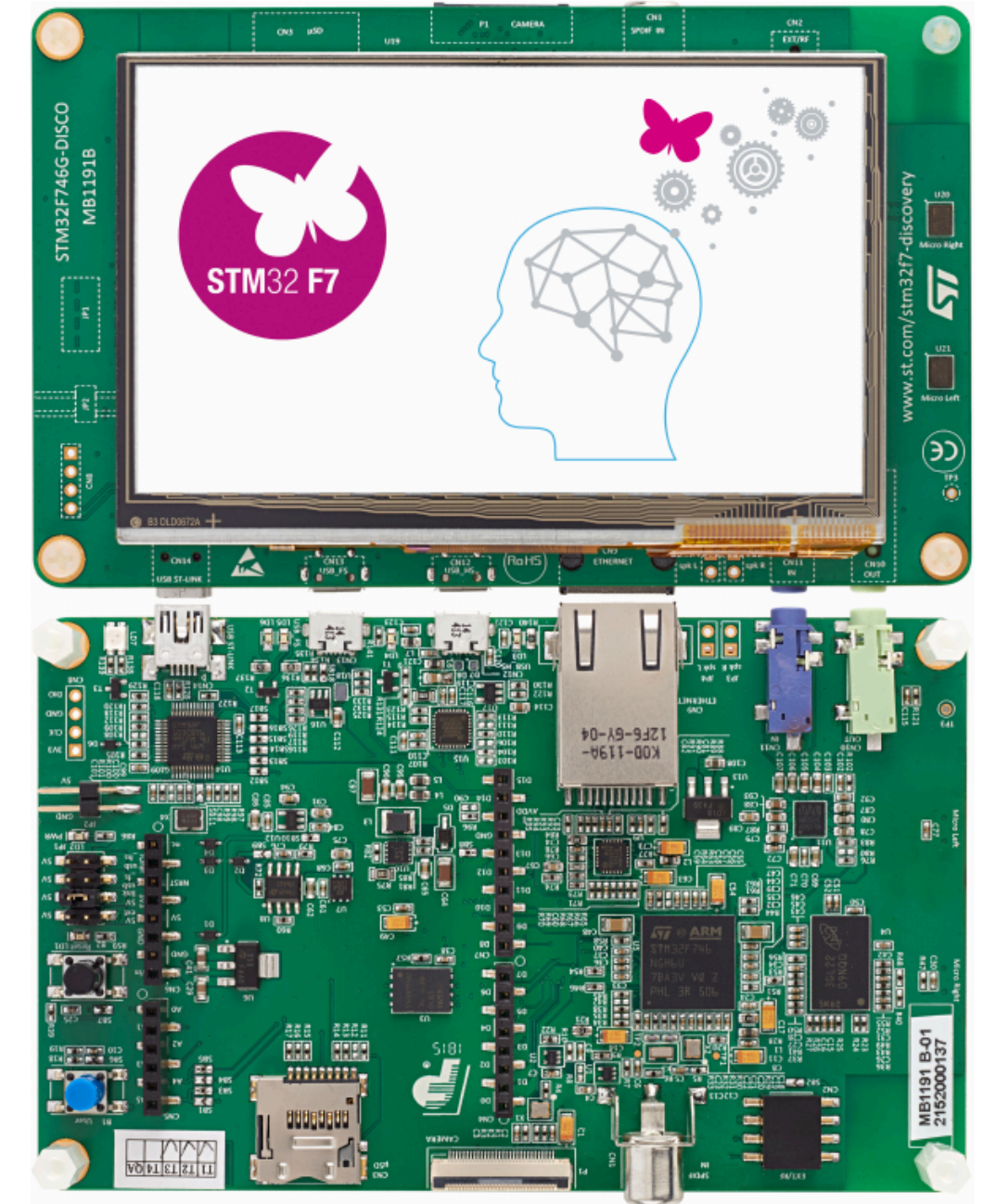
Empty

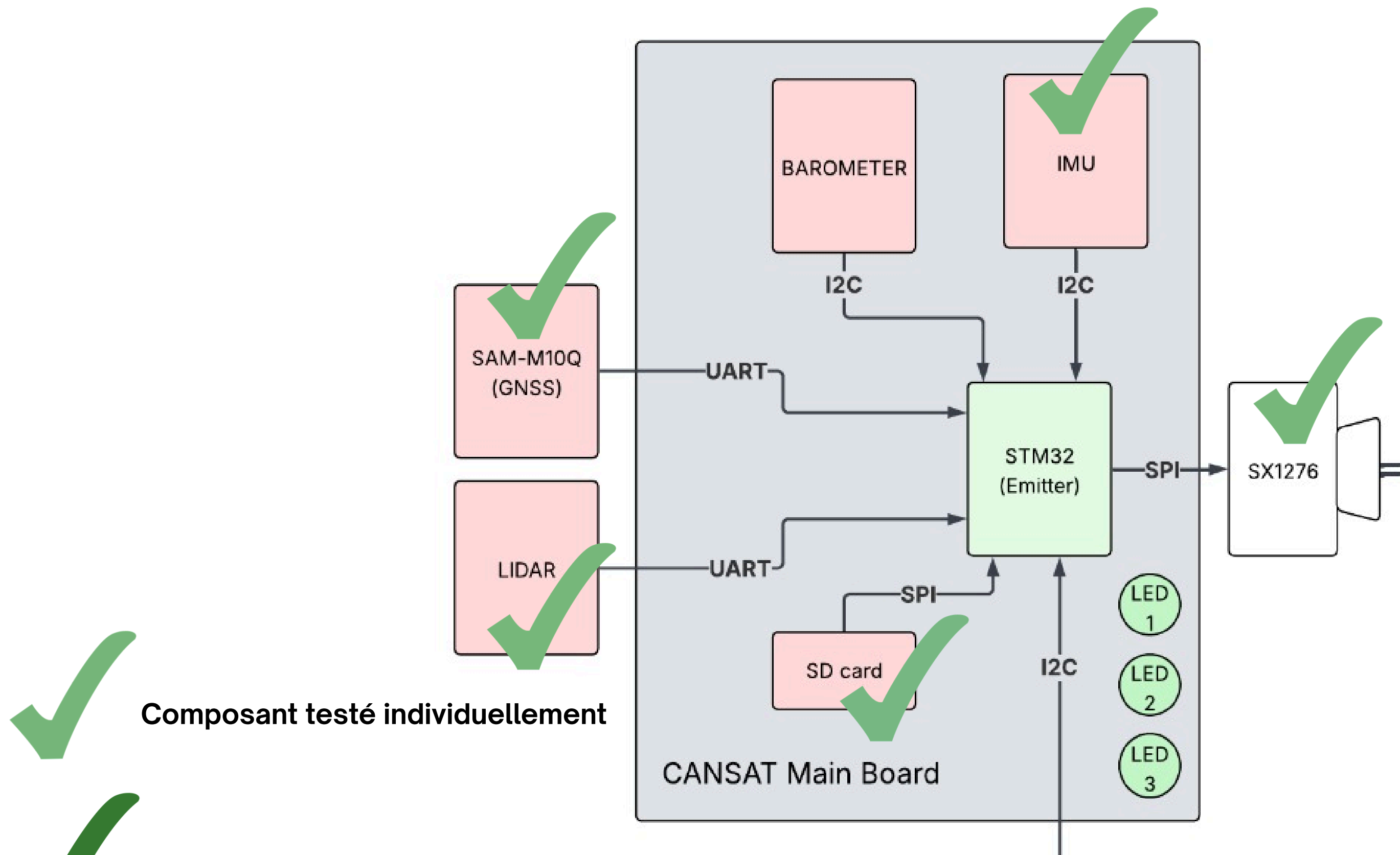
==== SD Card ====
TEST.TXT
Here !
Empty

Empty

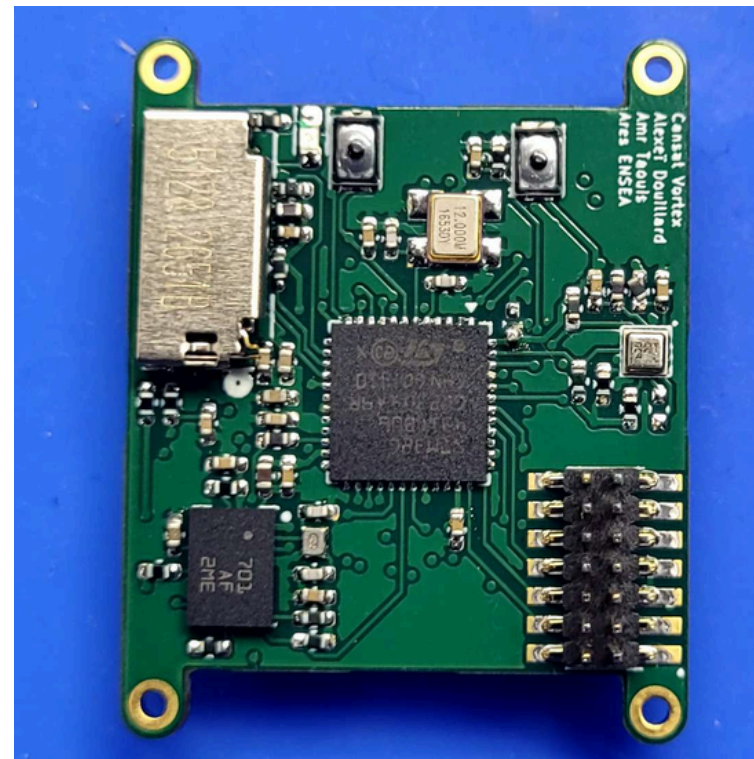
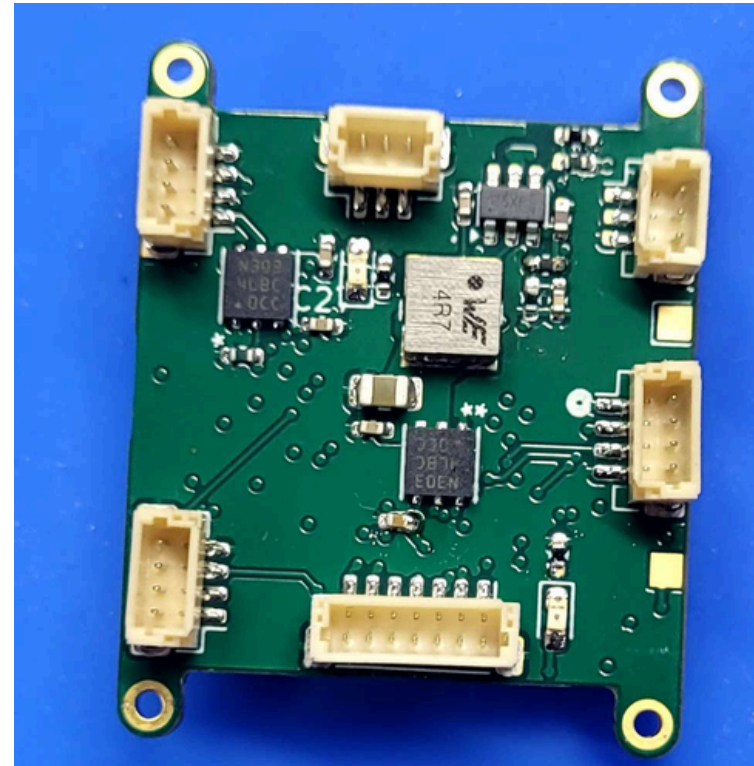
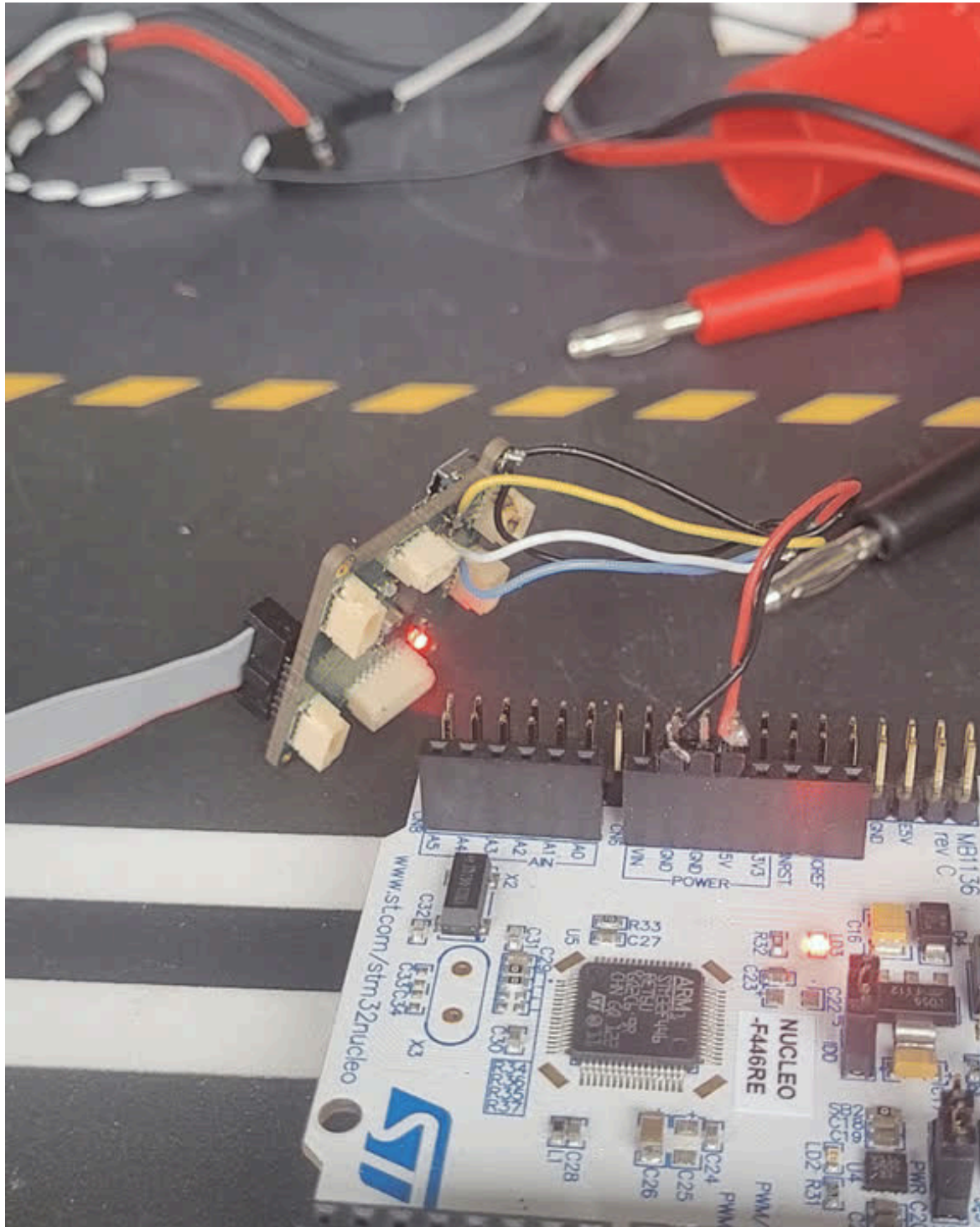
==== SD Card ====
TEST.TXT
Here !
Empty

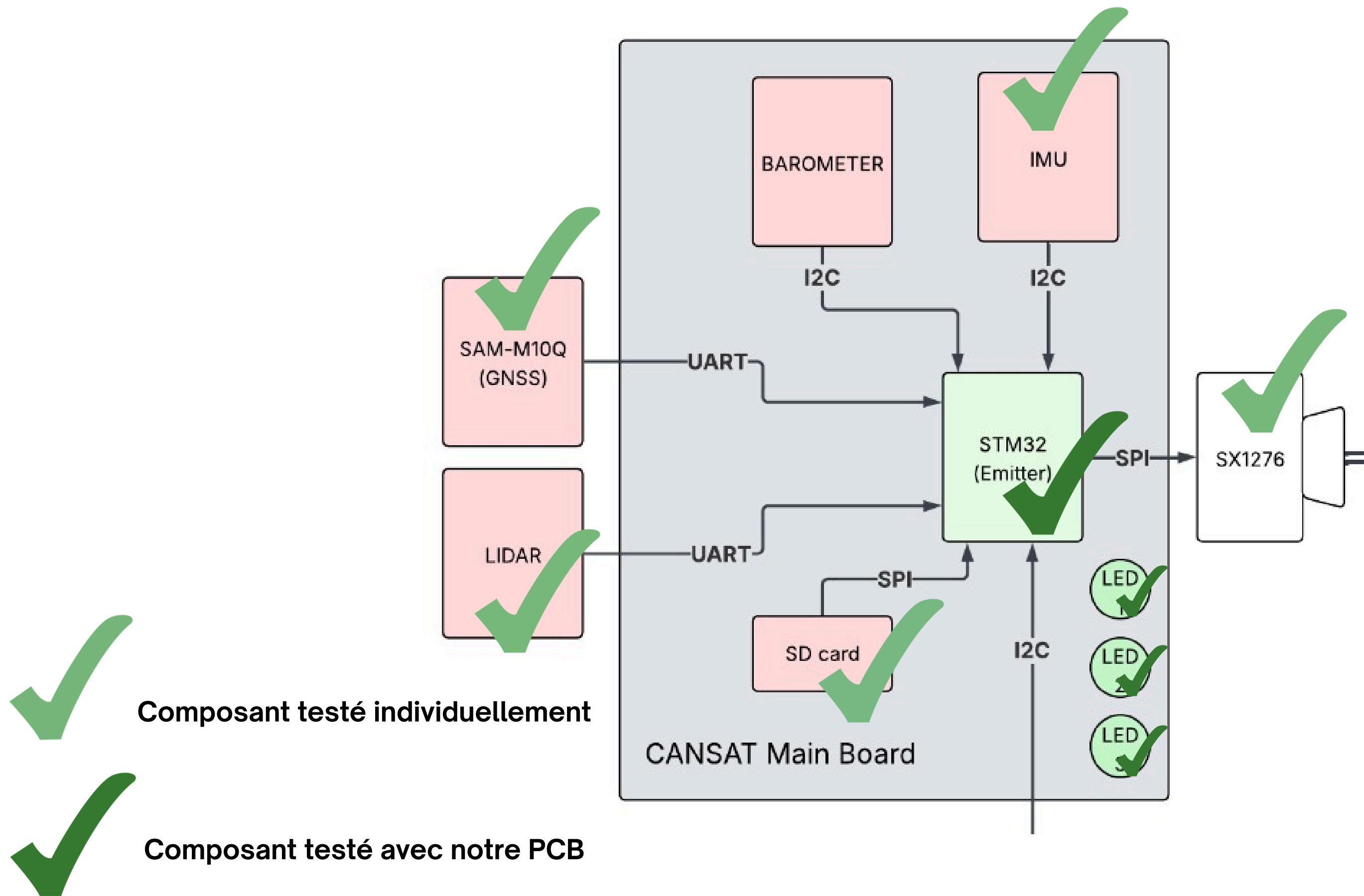
Hello World
```



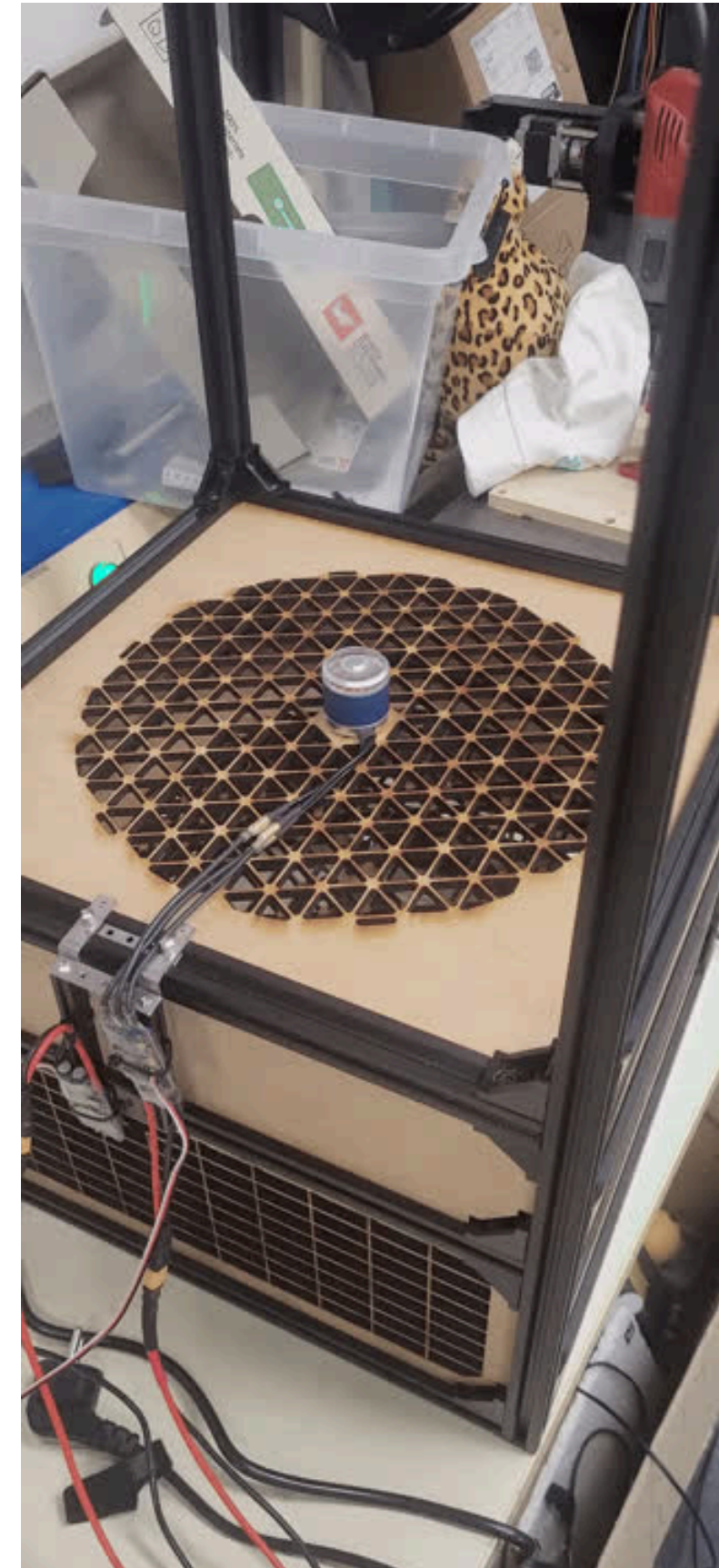
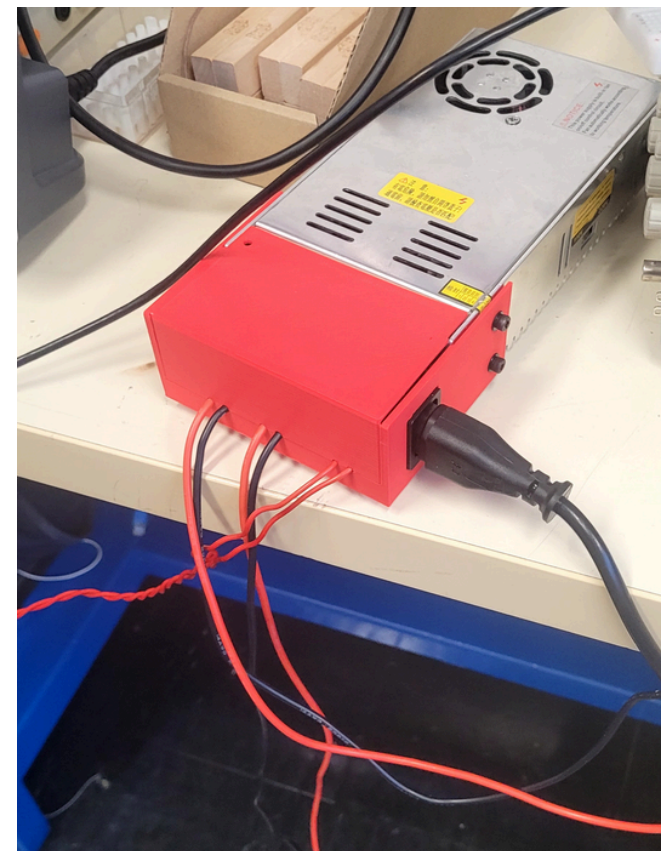
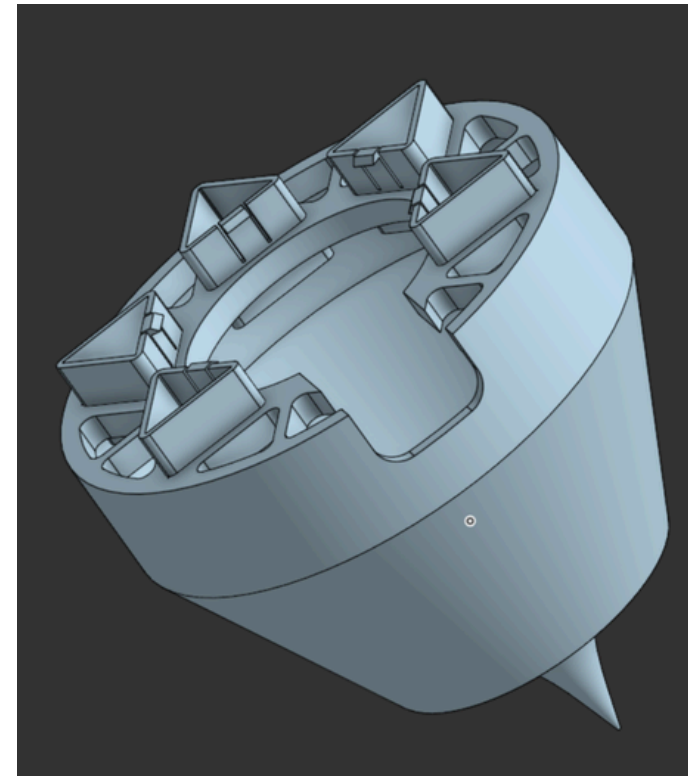
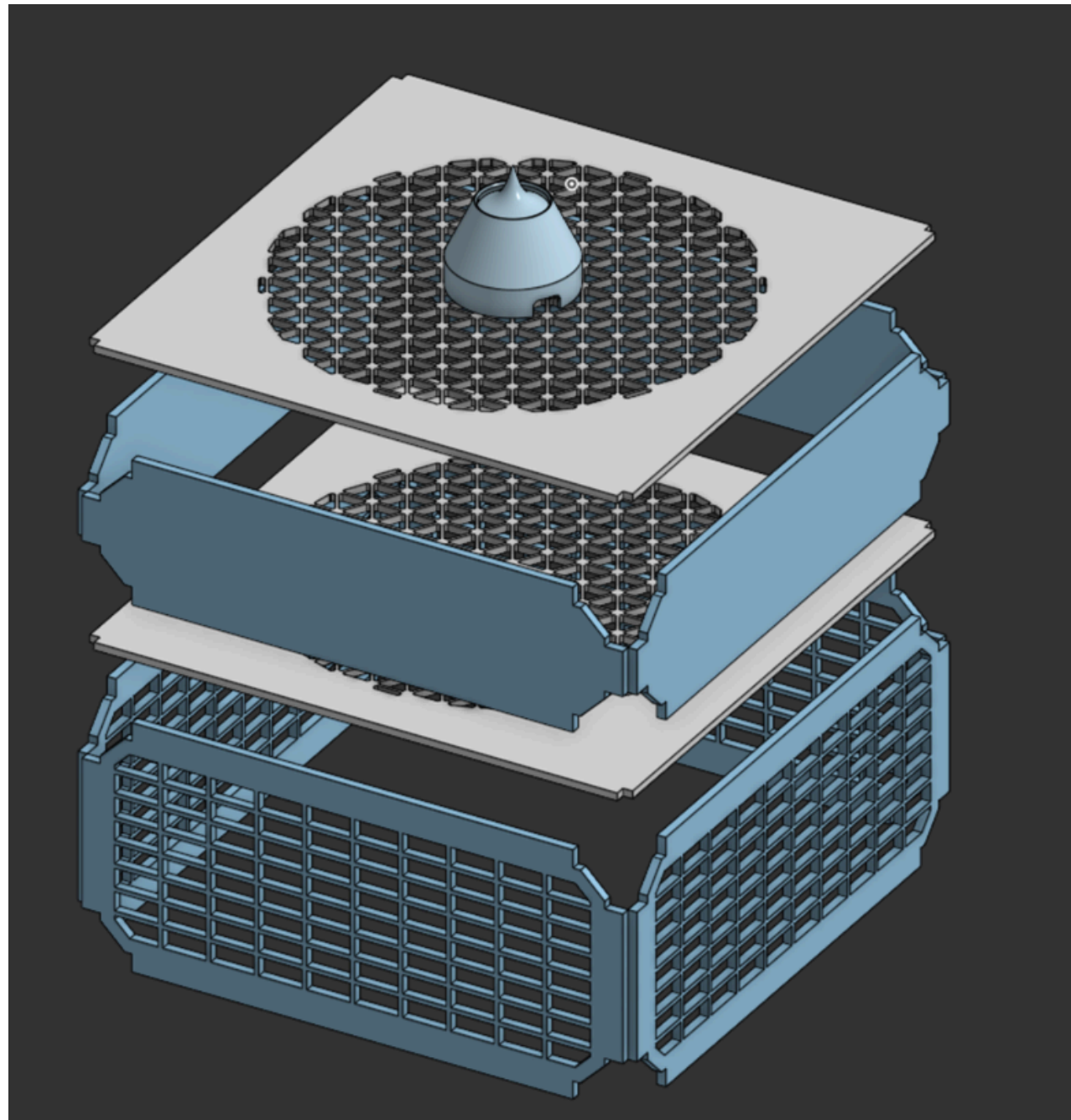


Test PCB



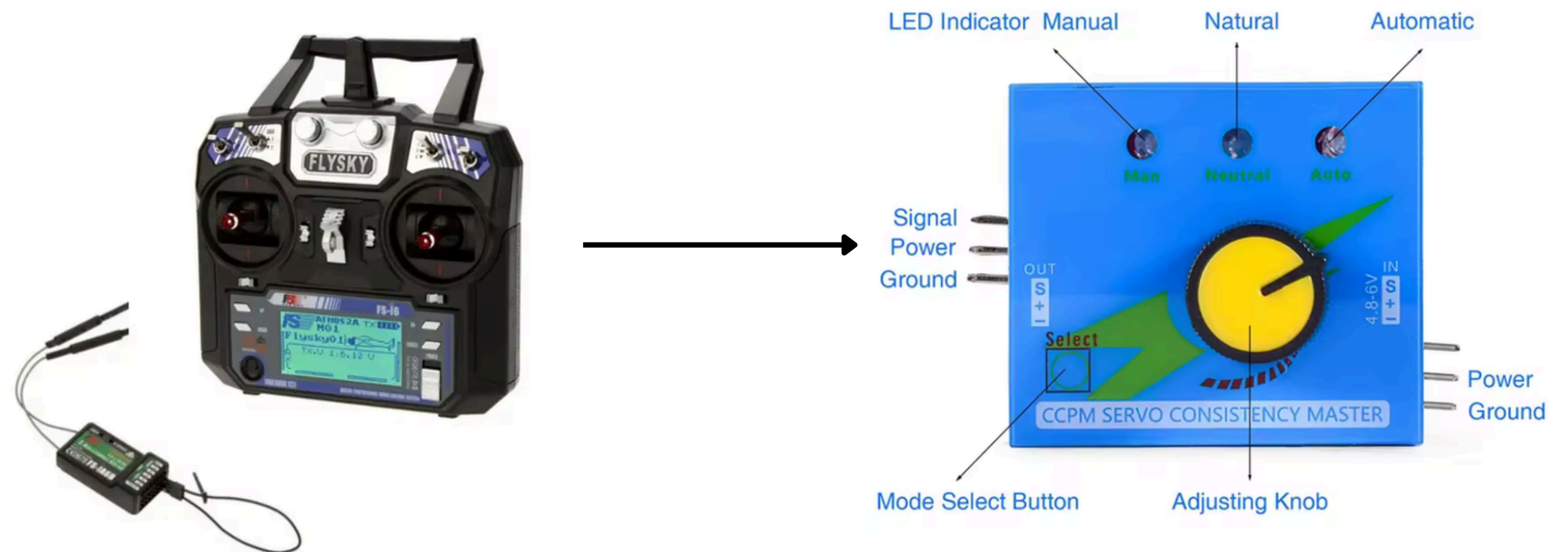


Soufflerie

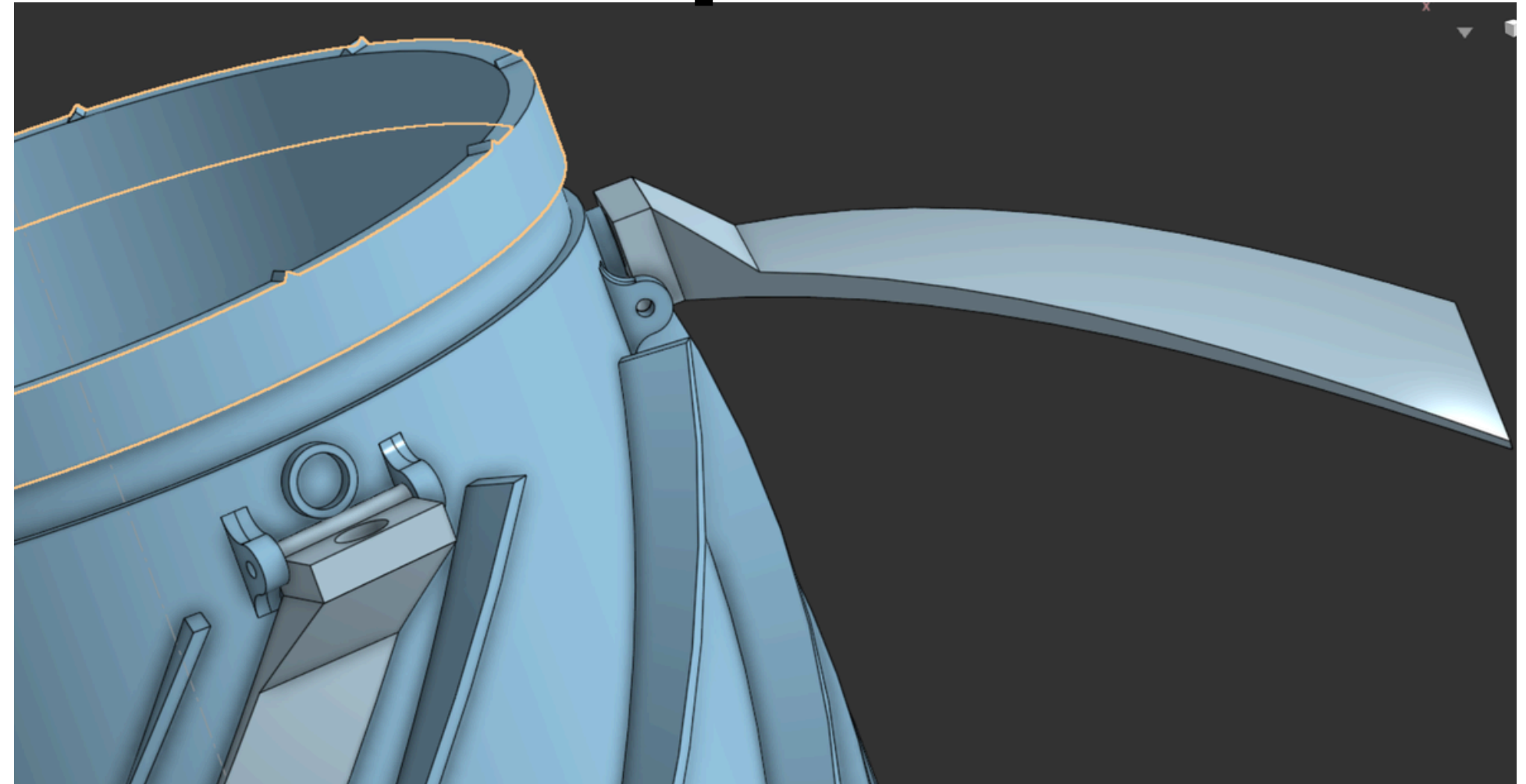
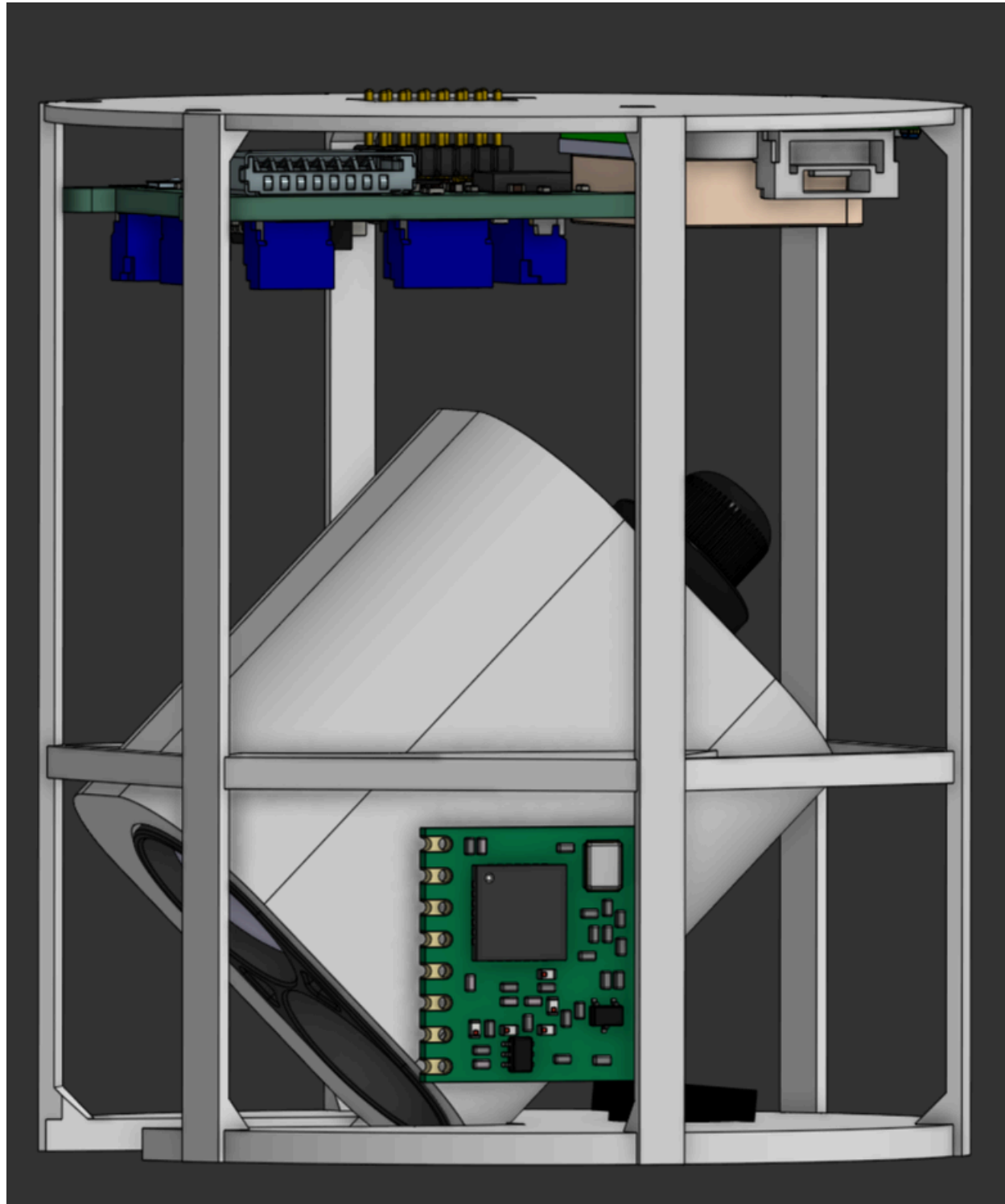


Ca rotationne !

Amélioration du contrôle des prochains test :



Avancement mécanique :



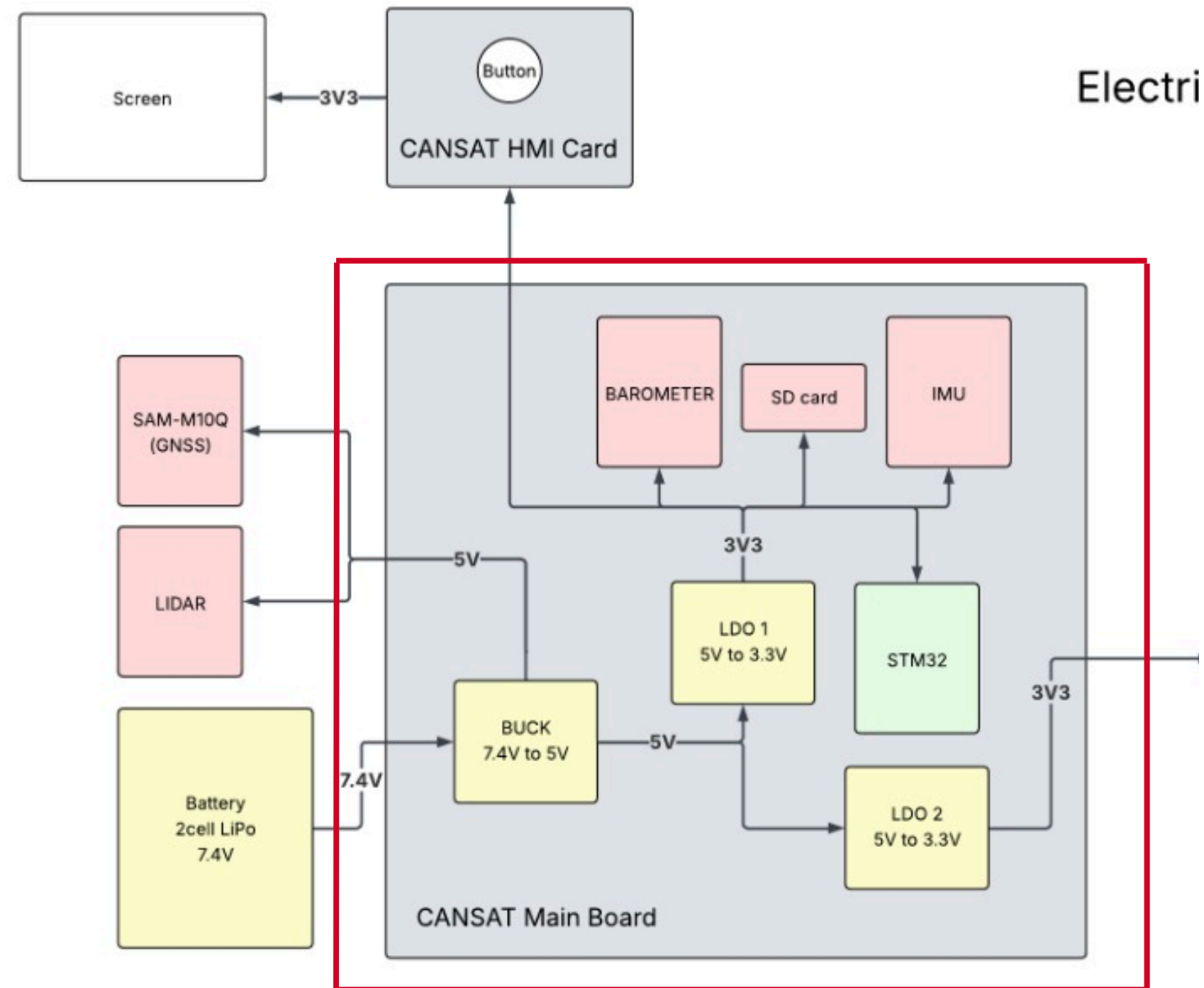
Progrès de la station au sol

```
⊗ (venv) hugo@cansat2425:~/lora_test $ /home/hugo/lora_test/venv/bin/python /home/hugo/lora_test/receiver.py
Listening for messages at 869.53 MHz (SF7, BW125, CR4/5)...

RxDone Interrupt detected.
Received Data: J
îâò|7D]gÎ&GÌ:gÀq7(TÛ_.SÄµdzC;.£|ýæs·Î·£}x'#õûiOæ\µÂÛ+âxL5ôÛõóÅ!;-ûÛ¼
RSSI: -106 dBm
SNR: 11.5 dB

RxDone Interrupt detected.
Received Data: CDNSAT FK
RSSI: -102 dBm
SNR: 8.0 dB
```

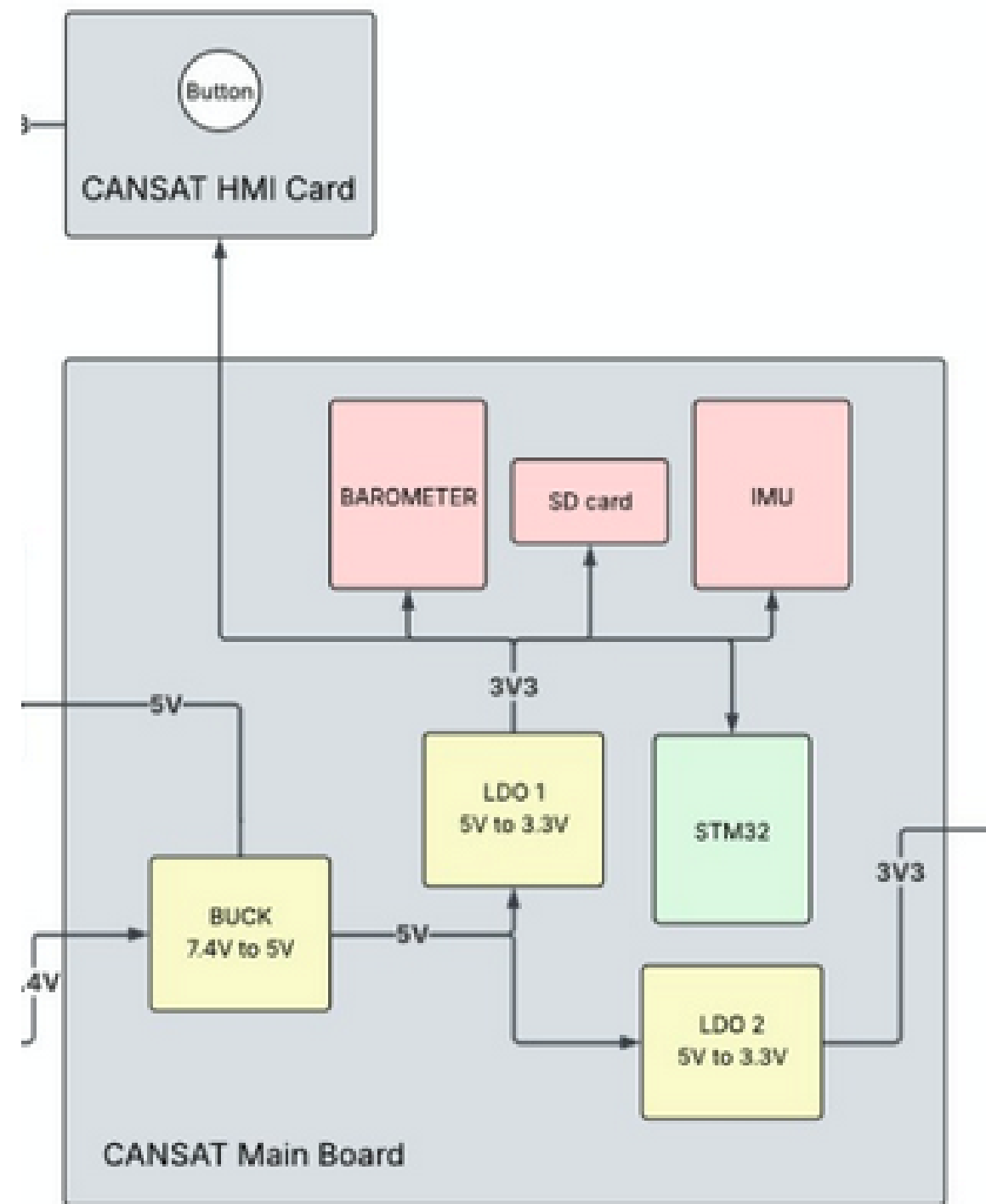

Progrès du Hardware



Mainboard V2

- Empreinte Quartz, buck (alimentation primaire 5V)
- Inversion d'alimentation GPS/LoRA (5V / 3.3VEN)
- Ajout d'un connecteur 6 pins pour l'IHM
- Sortie pour activer la caméra (EN)
- Sérigraphie

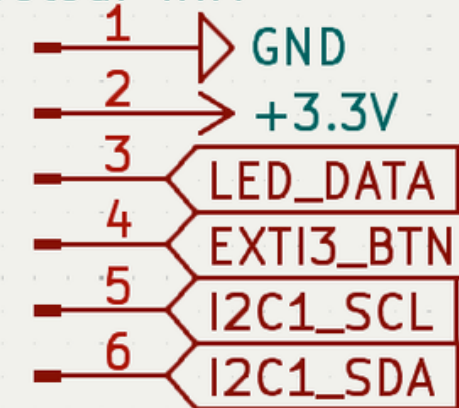
Mainboard V2



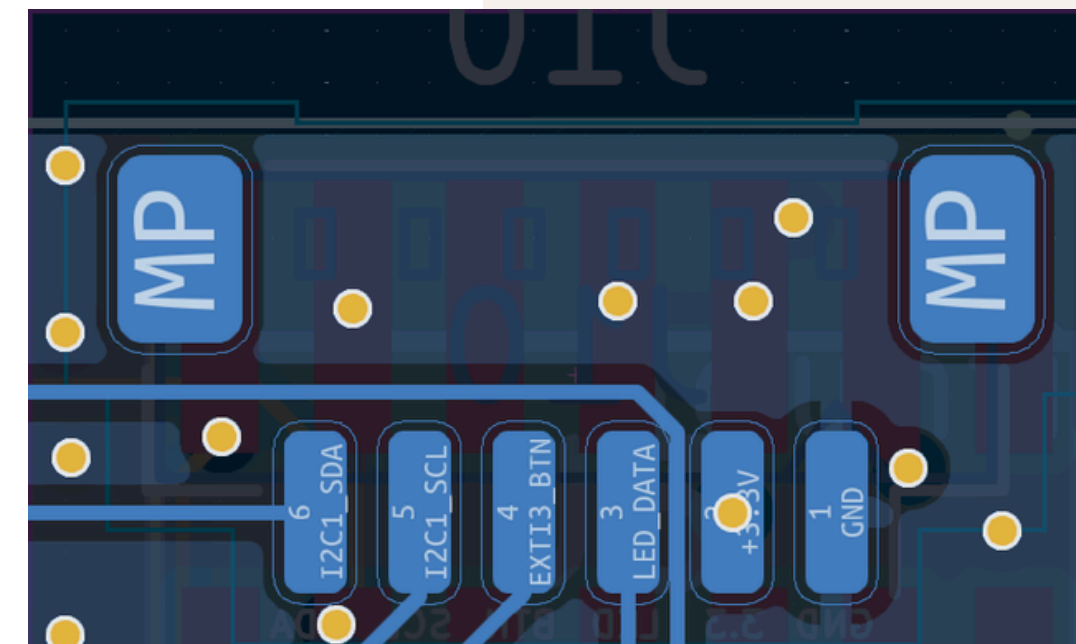
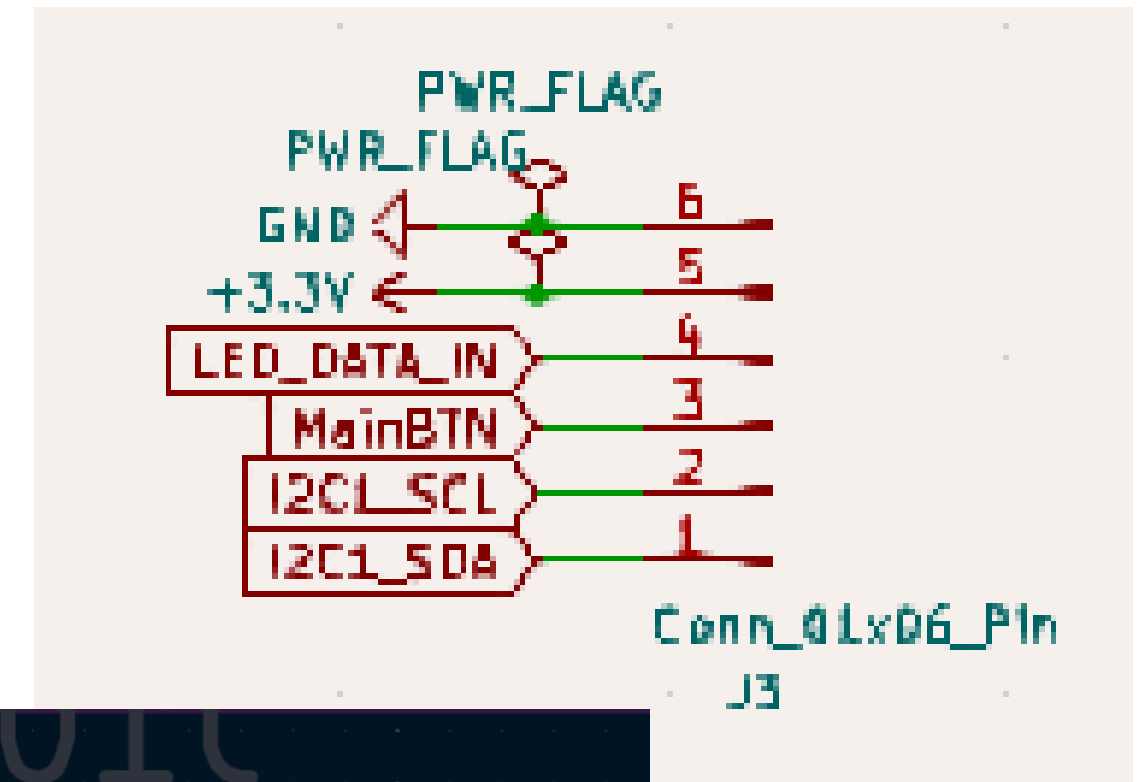
MainboardV2

Connecteurs IHM

J10
Connecteur IHM

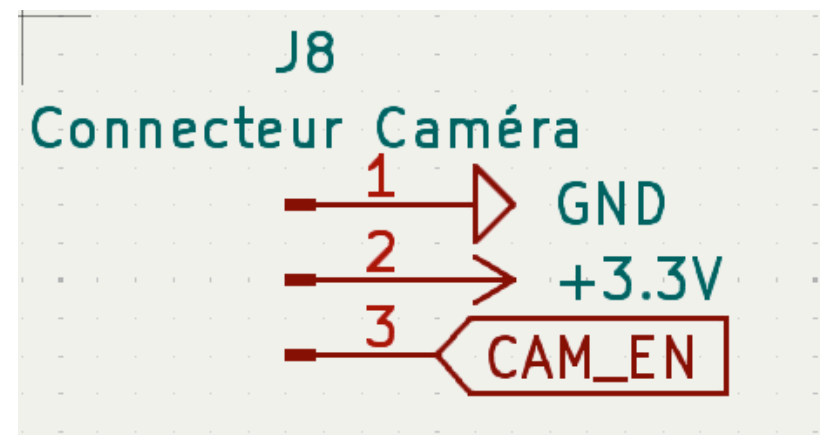
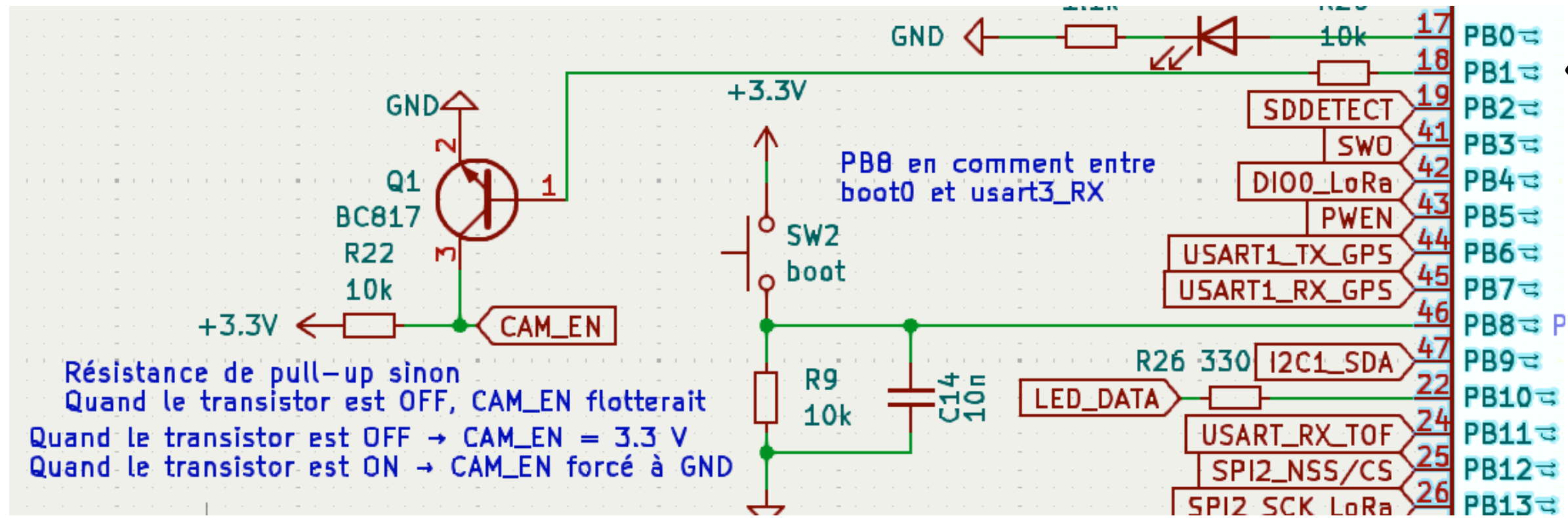


IHM



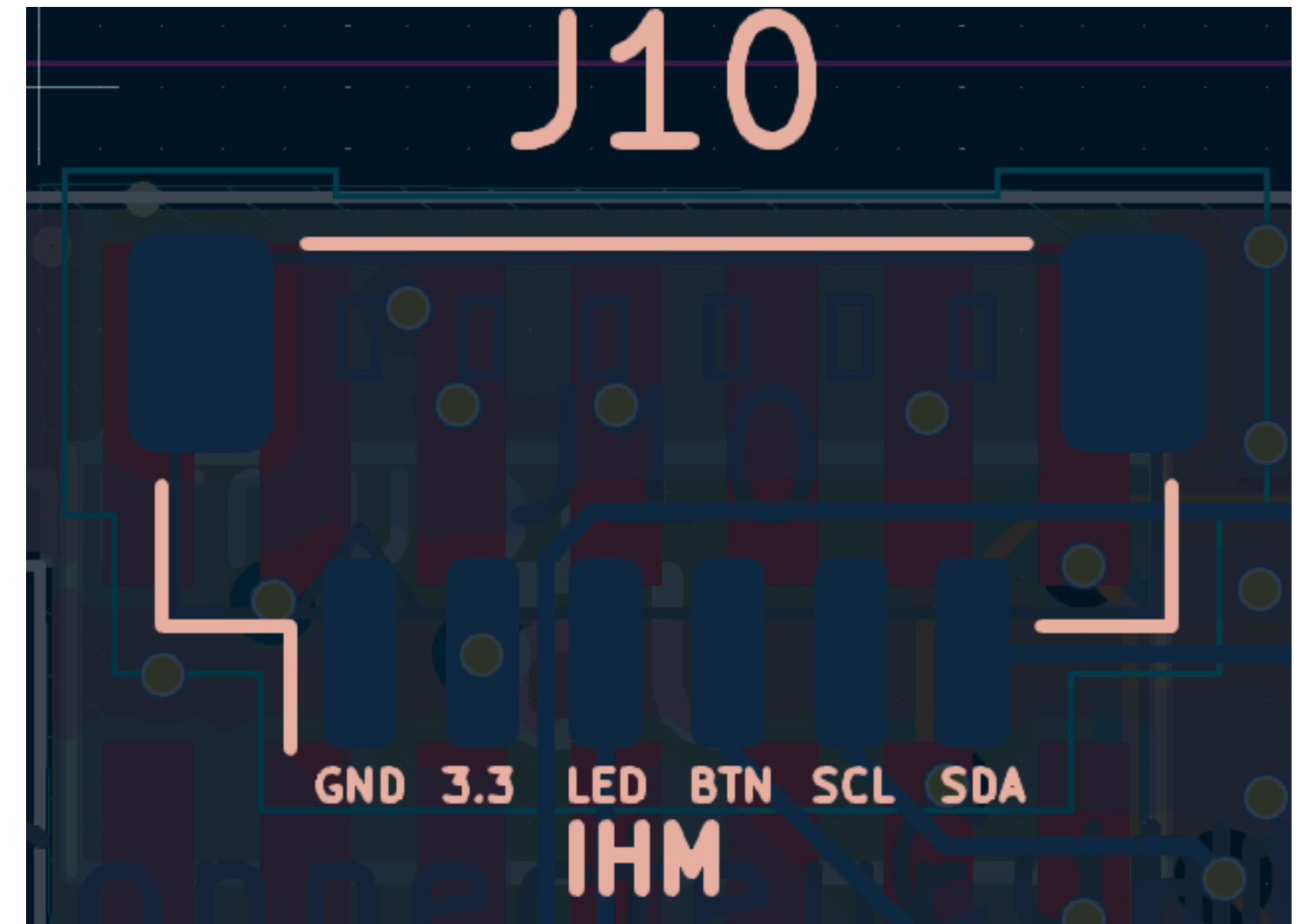
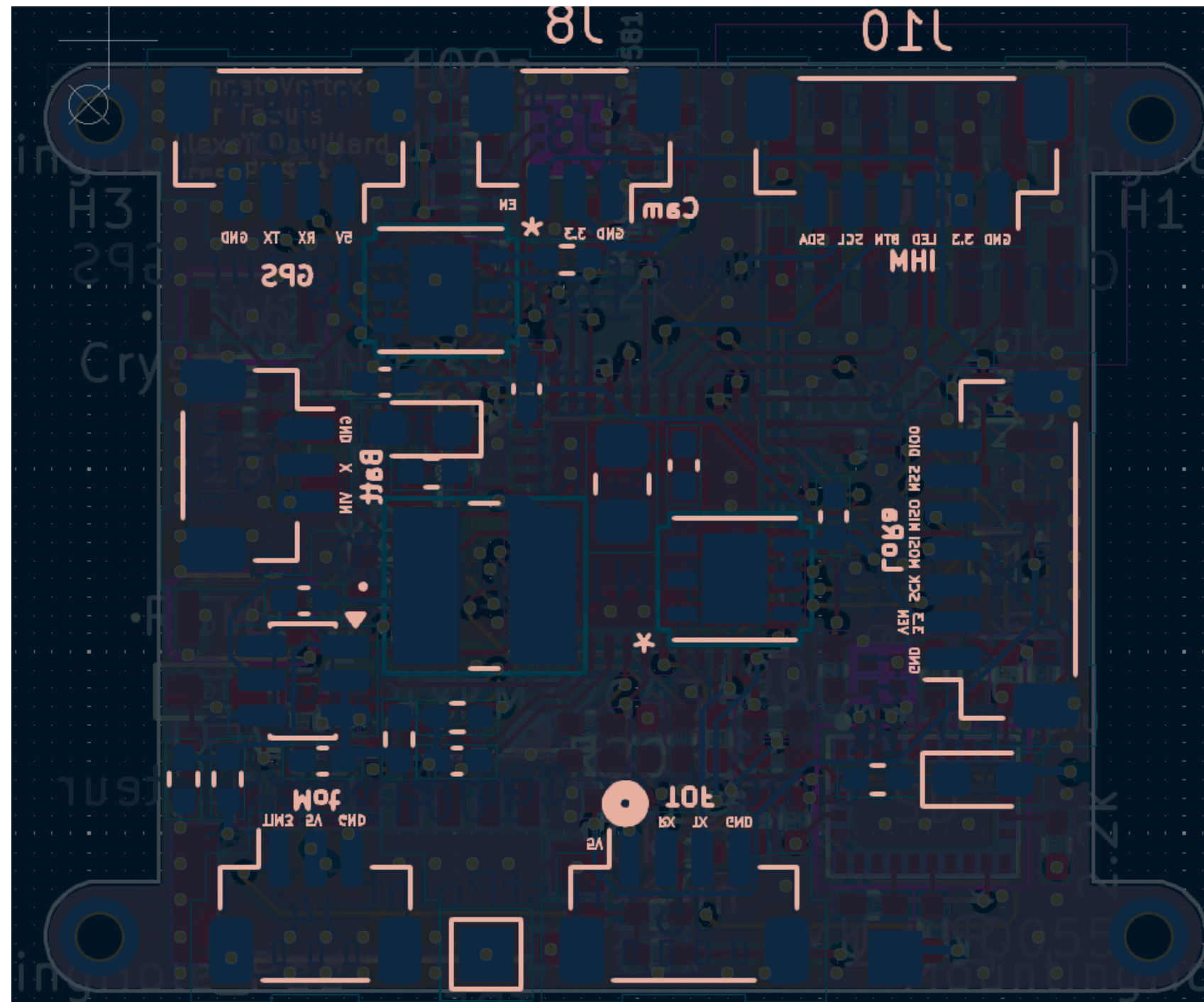
Mainboard V2

- Sortie pour activer la caméra (EN)

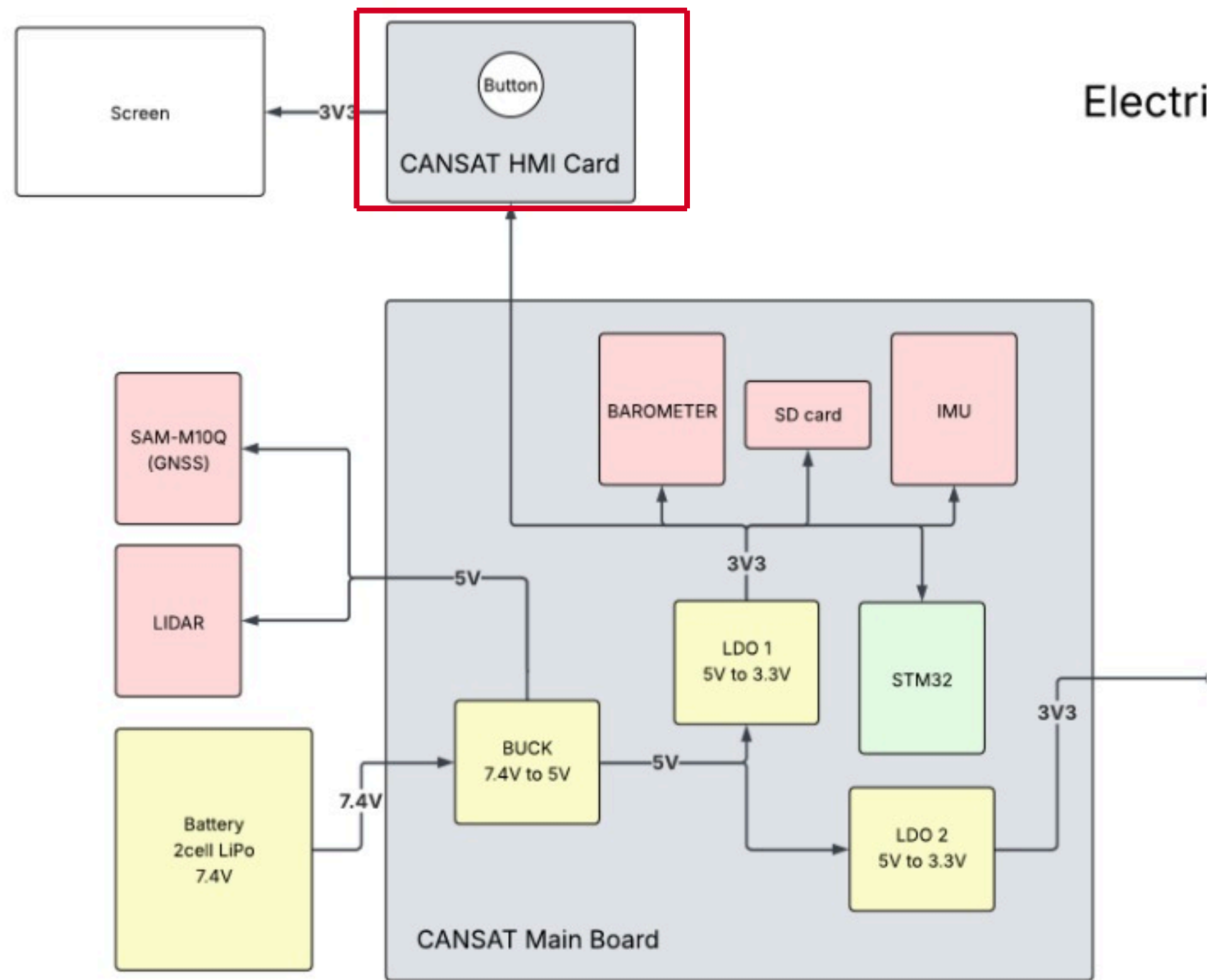


Mainboard V2

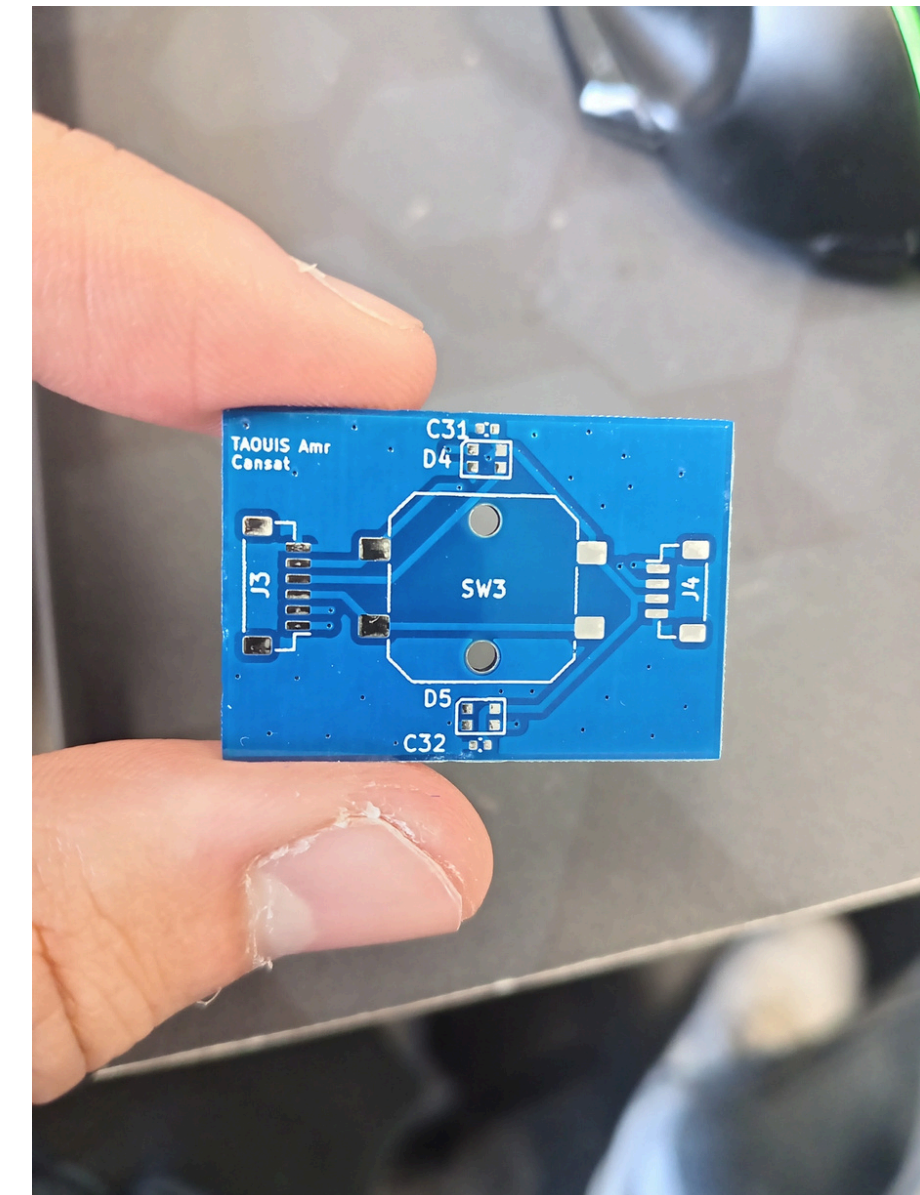
- Sérigraphie



Progrès du Hardware

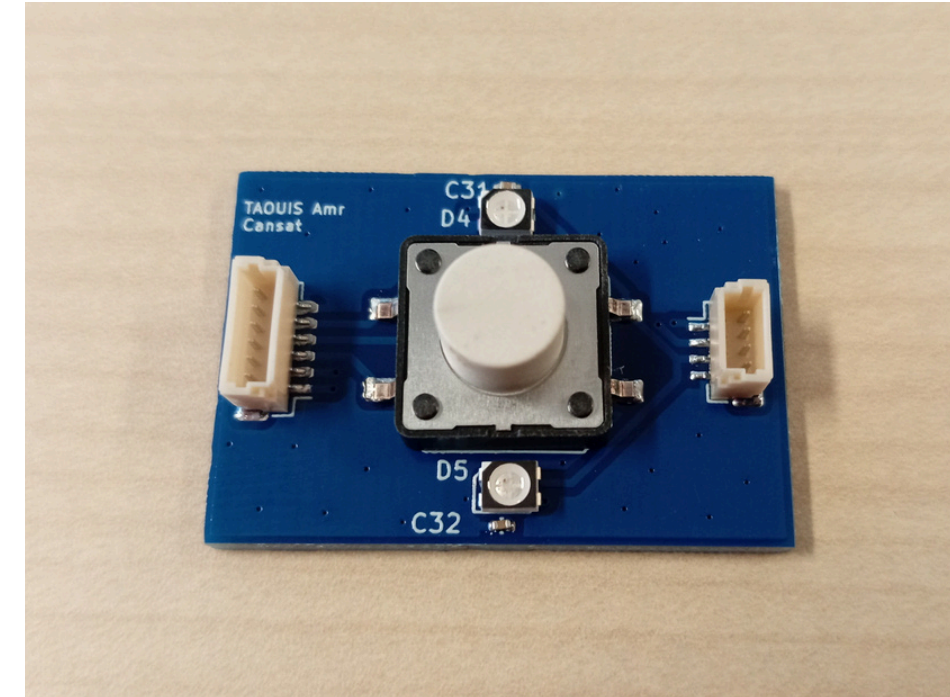
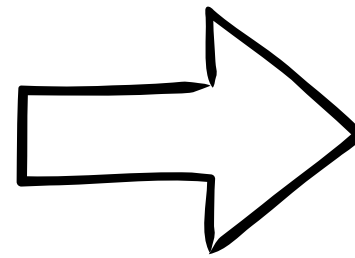
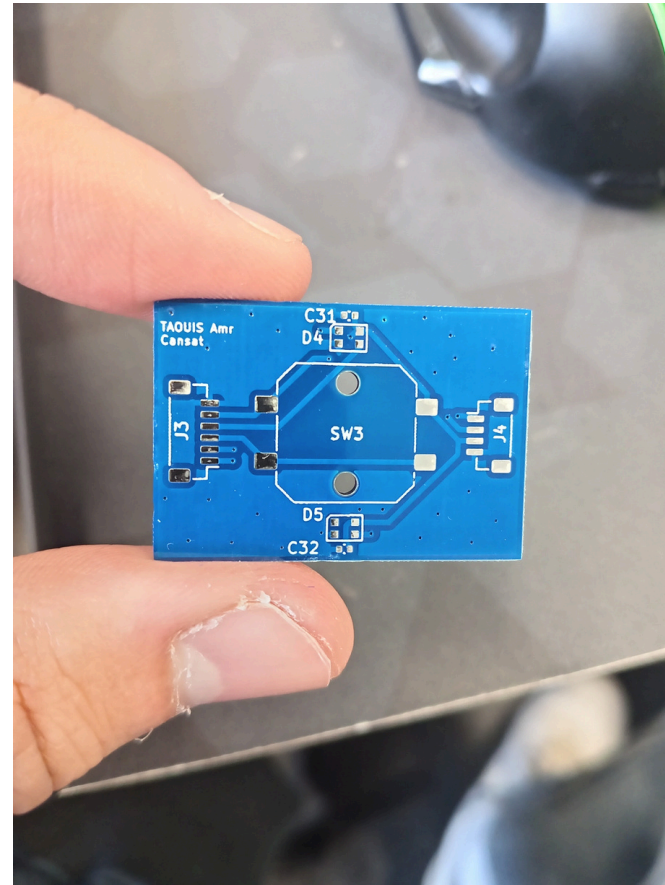


PCB IHM

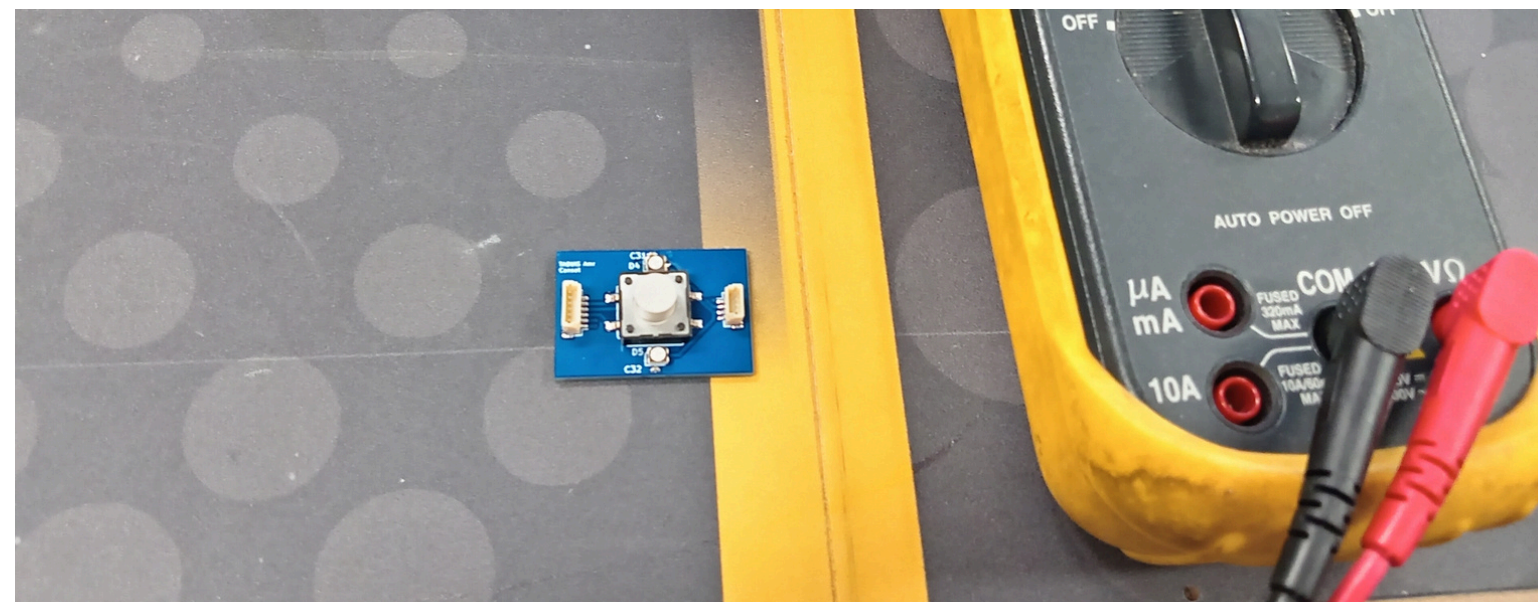


Progrès du Hardware

PCB IHM



Test



Problème potentiel :



LED à tester

Merci

